

Combining Services And Permission Theory TASK – 4

- 1. Create a new directory called mysite in your home folder, place an index.html file inside it, and set permissions so only your user can read and access the site files.**

Ans :

- # Create the directory
- `mkdir ~/mysite`

- # Create the index.html file
- `touch ~/mysite/index.html`

- # (Optional) Add some content
- `echo "<h1>Welcome to My Site</h1>" > ~/mysite/index.html`

- # Set ownership to your user
- `chown -R $(whoami):$(whoami) ~/mysite`

- # Set directory permissions:
- # Owner: rwx
- # Group: ---
- # Others: ---
- `chmod 700 ~/mysite`

- # Set file permissions:
- # Owner: rw-
- # Group: ---
- # Others: ---
- `chmod 600 ~/mysite/index.html`

2. Configure your Apache or Nginx server to serve the mysite directory as a website, then test if you can access it from your browser while logged in as your user.

Ans :

Option 1: Apache

- `mkdir -p ~/mysite`
- `echo "<h1>Welcome to My Site</h1>" > ~/mysite/index.html`
- `sudo nano /etc/apache2/sites-available/mysite.conf`
- `<VirtualHost *:80>`
- `ServerName localhost`
- `DocumentRoot /home/yourusername/mysite`
- `<Directory /home/yourusername/mysite>`
- `Options Indexes FollowSymLinks`
- `AllowOverride All`
- `Require all granted`
- `</Directory>`
- `</VirtualHost>`
- `sudo a2ensite mysite.conf`
- `sudo systemctl reload apache2`

Option 2: Nginx

- `sudo apt update`
- `sudo apt install nginx`

- `mkdir -p ~/mysite`
- `echo "<h1>Welcome to My Site</h1>" > ~/mysite/index.html`

- `sudo nano /etc/nginx/sites-available/mysite`

- `server {`
- `listen 80;`
- `server_name localhost;`

- `root /home/yourusername/mysite;`
- `index index.html;`

- `location / {`
- `try_files $uri $uri/ =404;`
- `}`
- `}`

- `sudo ln -s /etc/nginx/sites-available/mysite /etc/nginx/sites-enabled/`
- `sudo nginx -t`
- `sudo systemctl reload nginx`

- `chmod o+x /home/yourusername`
- `chmod -R o+r ~/mysite`

**3. Change the permissions of the mysite directory so that another user on the system cannot access the site files directly. Try to access the files as the other user and note the result.

Hint: Use chmod and su or a different terminal window for the second user.**

Ans :

➤ Change the permissions

- chmod 700 ~/mysite
- chmod 600 ~/mysite/*

➤ Example output:

- drwx----- 2 student student 4096 Jun 16 14:00 mysite
- -rw----- 1 student student 29 Jun 16 14:00 index.html

➤ Switch to another user

- su – otheruser

➤ Attempt to access the files

- ls /home/student/mysite

➤ Expected result

- ls: cannot open directory '/home/student/mysite': Permission denied

4. Document the steps you took to restrict access to your hosted site, including the commands used, permission settings, and

screenshots or outputs showing successful and failed access attempts.

Ans :

Step 1: Verify Current Permissions

I checked the existing permissions of the website directory and files using:

- `ls -ld ~/mysite`
- `ls -l ~/mysite`
- Example output:
 - `drwxr-xr-x 2 student student 4096 Jun 16 14:00 mysite`
 - `-rw-r--r-- 1 student student 29 Jun 16 14:00 index.html`

The output showed that other users had read and execute permissions.

Step 2: Restrict Access

I modified the permissions so that only the owner could access the directory and files.

- Commands used:
 - `chmod 700 ~/mysite`
 - `chmod 600 ~/mysite/*`
- Permission settings:
 - Directory (mysite): 700
 - Owner: read, write, execute
 - Group: no access
 - Others: no access

➤ Files (index.html): 600

- Owner: read, write
- Group: no access
- Others: no access

Step 3: Verify Updated Permissions

I confirmed the changes using:

- `ls -ld ~/mysite`
- `ls -l ~/mysite`

Output:

- `drwx----- 2 student student 4096 Jun 16 14:15 mysite`
- `-rw----- 1 student student 29 Jun 16 14:15 index.html`

Step 4: Test Access as Another User

I switched to another user account:

- `su - otheruser`

Then attempted to access the website files:

- `ls /home/student/mysite`

Output:

- `ls: cannot open directory '/home/student/mysite': Permission denied`

Next, I attempted to read the website file:

- `cat /home/student/mysite/index.html`

Output:

- `cat: /home/student/mysite/index.html: Permission denied`

Results

- Successful Access (Owner)
- `cat ~/mysite/index.html`

Output:

- `<h1>Welcome to My Site</h1>`
- Failed Access (Other User)
- `cat /home/student/mysite/index.html`

Output:

- Permission denied